Implementation of Model Predictive Control (MPC) for a Coupled Nonlinear MIMO Plant

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Resumen

This paper presents the design and implementation of a Model Predictive Control (MPC) strategy for a nonlinear coupled MIMO system. The approach is based on a reduced-order linear model derived from LSTM neural network identification. The nonlinear system dynamics were first identified using an LSTM network trained from input–output data. The identified model was linearized around an operating point and reduced to a discrete-time state-space representation suitable for MPC design.

The reduced-order model is used to formulate a constrained linear MPC controller capable of handling multivariable interactions and operational constraints. Simulation results demonstrate accurate reference tracking, constraint satisfaction, and stable closed-loop behavior. The results are compared against traditional PID and simple MPC controllers, highlighting the advantages of the proposed method.

Model Predictive Control LSTM Neural Networks Nonlinear Systems MIMO Systems Model Reduction

1. Introduction

Nonlinear multivariable systems are common in industrial applications, where coupling effects, nonlinearities, and operational constraints complicate controller design. Accurate first-principles modeling is often challenging, motivating the use of data-driven approaches.

Long Short-Term Memory (LSTM) neural networks have proven effective in capturing nonlinear system dynamics from measured data. However, neural network models are often computationally heavy for direct use in control and lack interpretability.

In this work, the nonlinear MIMO system is first identified using LSTM networks. The model is linearized around a nominal operating point, and a reduced-order linear discrete-time model is obtained for MPC design. This paper focuses on the design, implementation, and simulation of the MPC controller.

1.1. Motivation and Objectives

The main objective is to design an MPC that ensures accurate reference tracking while satisfying operational constraints. Using a linearized reduced-order model allows efficient optimization and real-time implementation, even for originally nonlinear systems.

1.2. Paper Structure

The paper is organized as follows: Section 2 details the system modeling and identification; Section 3 introduces MPC fundamentals; Section 4 presents the MPC design and implementation; Section 5 discusses simulation results; Section 6 provides analysis and discussion; Section 7 presents conclusions and future work.

2. System Modeling and Identification

2.1. Nonlinear MIMO System

The plant under consideration has two inputs and two outputs. Denoting the inputs as $u_1(t)$ and $u_2(t)$ and the outputs as $y_1(t)$ and $y_2(t)$, the nonlinear dynamics are described by:

$$\frac{dy_1(t)}{dt} = f_1(y_1, y_2, u_1, u_2), \quad \frac{dy_2(t)}{dt} = f_2(y_1, y_2, u_1, u_2) \tag{1}$$

where f_1 and f_2 are nonlinear functions capturing system interactions.

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2.2. LSTM-based Identification

Historical input-output data is used to train an LSTM neural network to approximate the nonlinear dynamics:

$$y(t+1) = \mathcal{F}_{\text{LSTM}}(y(t), u(t))$$

The trained model is then linearized around a nominal operating point (x_0, u_0) using a first-order Taylor expansion.

2.3. Reduced-order Linear Model

The linearized discrete-time state-space model is:

$$x(k+1) = Ax(k) + Bu(k), \quad y(k) = Cx(k) \tag{2}$$

with $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^2$, $y(k) \in \mathbb{R}^2$, and A, B, C reduced-order matrices. The model captures the dominant system dynamics and is suitable for MPC design.

3. Fundamentals of Model Predictive Control (MPC)

3.1. MPC Concepts

MPC predicts future system behavior over a prediction horizon N_p and solves an optimization problem to minimize a cost function:

$$J = \sum_{i=1}^{N_p} \|y(k+i|k) - r(k+i)\|_Q^2 + \sum_{i=0}^{N_c-1} \|\Delta u(k+i|k)\|_R^2 \tag{3}$$

where $r(k)$ is the reference trajectory, Q and R are weighting matrices, $\Delta u(k) = u(k) - u(k-1)$, and N_c is the control horizon.

3.2. Advantages of MPC

- Explicit handling of constraints on inputs and outputs.
- Improved tracking performance via anticipation of future system behavior.
- Effective management of MIMO interactions.

3.3. Constraints

$$u_{\text{mín}} \leq u(k) \leq u_{\text{máx}} \tag{4}$$

$$y_{\text{mín}} \leq y(k) \leq y_{\text{máx}} \tag{5}$$

4. MPC Design and Implementation

4.1. Prediction Model

The MPC uses the reduced-order linear discrete-time model for predictions.

4.2. Cost Function and Horizons

A quadratic cost function balances output tracking and control effort. Prediction and control horizons are tuned empirically, typically $N_p = 20$ and $N_c = 5$.

4.3. Implementation in MATLAB

The controller is implemented using MATLAB MPC Toolbox. Constraints and weighting matrices are included directly.

4.4. Evaluation Metrics

Performance is evaluated using:

- RMSE (Root Mean Square Error)
- MAE (Mean Absolute Error)
- IAE (Integral Absolute Error)

Cuadro 1: Comparison of control strategies

	Linearized MPC (robust)	Linearized MPC (simple)	PID MIMO
RMSE	0.442	0.445	0.581
MAE y1	0.346	0.354	0.489
MAE y2	0.453	0.454	0.424
IAE y1	34.61	35.43	48.92
IAE y2	45.30	45.35	42.40
Final Error y1/y2	0.230/0.257	0.238/0.258	0.364/0.137

5. Simulation Results

5.1. Reference Tracking

Simulations demonstrate accurate tracking of desired setpoints under varying reference trajectories.

5.2. Constraint Satisfaction

Input and output limits are never violated, showing that the MPC properly enforces constraints.

5.3. Comparison with Other Controllers

The linearized MPC outperforms simple MPC and PID controllers in terms of RMSE and IAE. Figures 1 and 2 illustrate the outputs and control signals.

tracking.png

Figura 1: Reference tracking of MPC controller.

control_signals.png

Figura 2: Control inputs applied by MPC.

6. Analysis and Discussion

The reduced-order MPC achieves:

- Stable closed-loop behavior.
- Smooth control actions respecting constraints.
- Better performance compared to baseline PID and simple MPC.

Limitations include potential performance degradation for operating points far from linearization.

7. Conclusion and Future Work

This work demonstrates that linearized MPC based on LSTM-identified models is effective for nonlinear coupled MIMO systems. Future work will explore:

- Robustness analysis under disturbances.
- Nonlinear MPC implementation using full LSTM models.
- Extension to higher-dimensional systems.

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